

Management of Geriatric patients

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Learning outcomes

At the end of the lecture student will be able to

- ▶ Understand why it is important to know geriatric patients
- ▶ Understand factors affecting oral health in geriatric patients
- ▶ List And describe common oral diseases and thier management
- ▶ Understand importance of preventive and palliative care in geriatrics
- ▶ Describe various prosthetic options for oral rehabilitation



Introduction



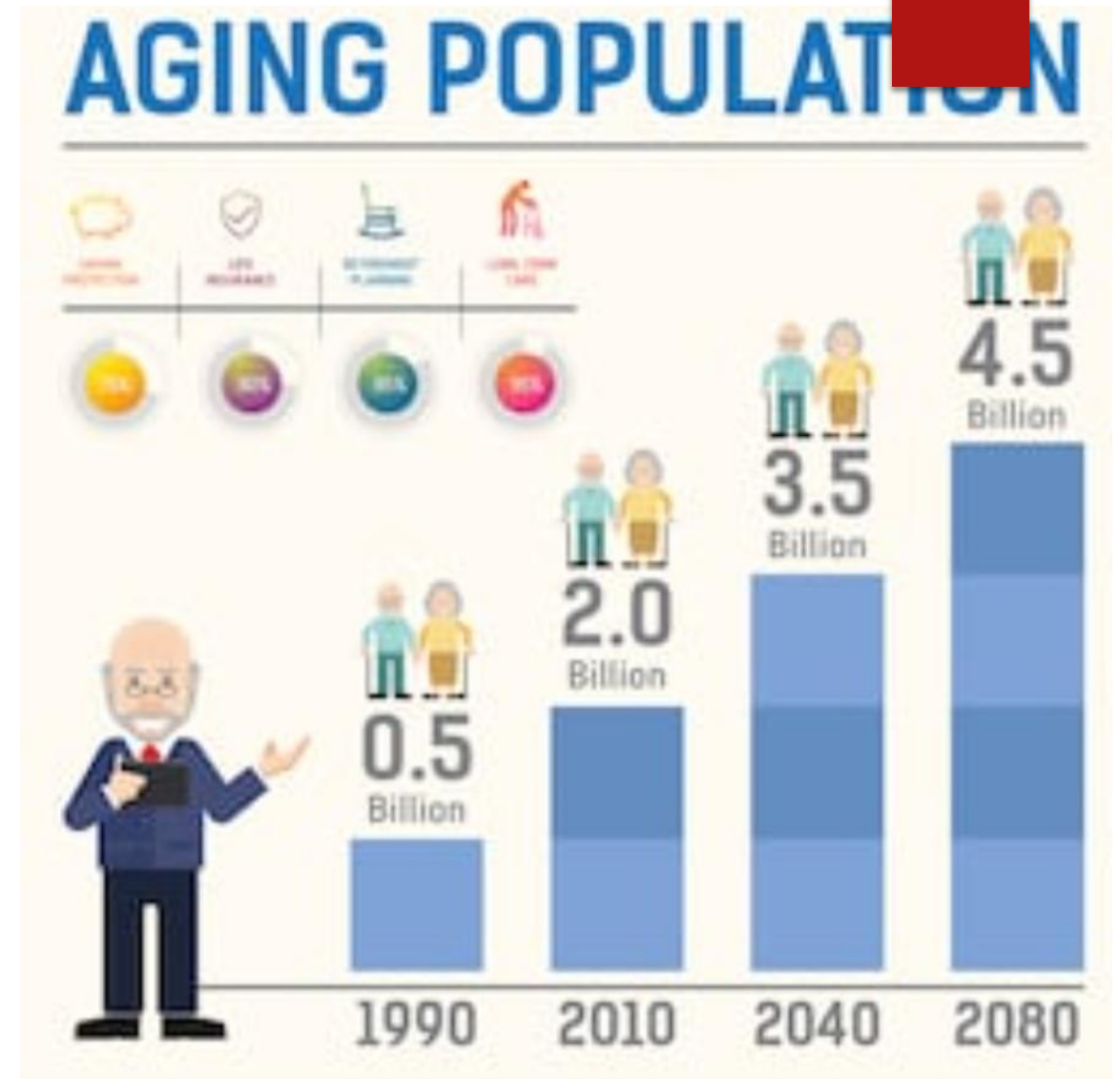
- ▶ Aging is a **natural process**. Old age should be regarded as a normal, inevitable biological phenomenon.
- ▶ Aging is a **progressive generalized impairment of function** resulting in a loss of adaptive response to stress and in a growing risk of age-associated disease.
- ▶ As a result of the advances made in medicine and public health measures in the last half of the 20th century, there is a **substantial increase in the life span** of man.
- ▶ With more people living to older ages and the older population getting older. This demographic change will have a major **impact on the delivery of general and oral-health care**, as well as on the providers of these services

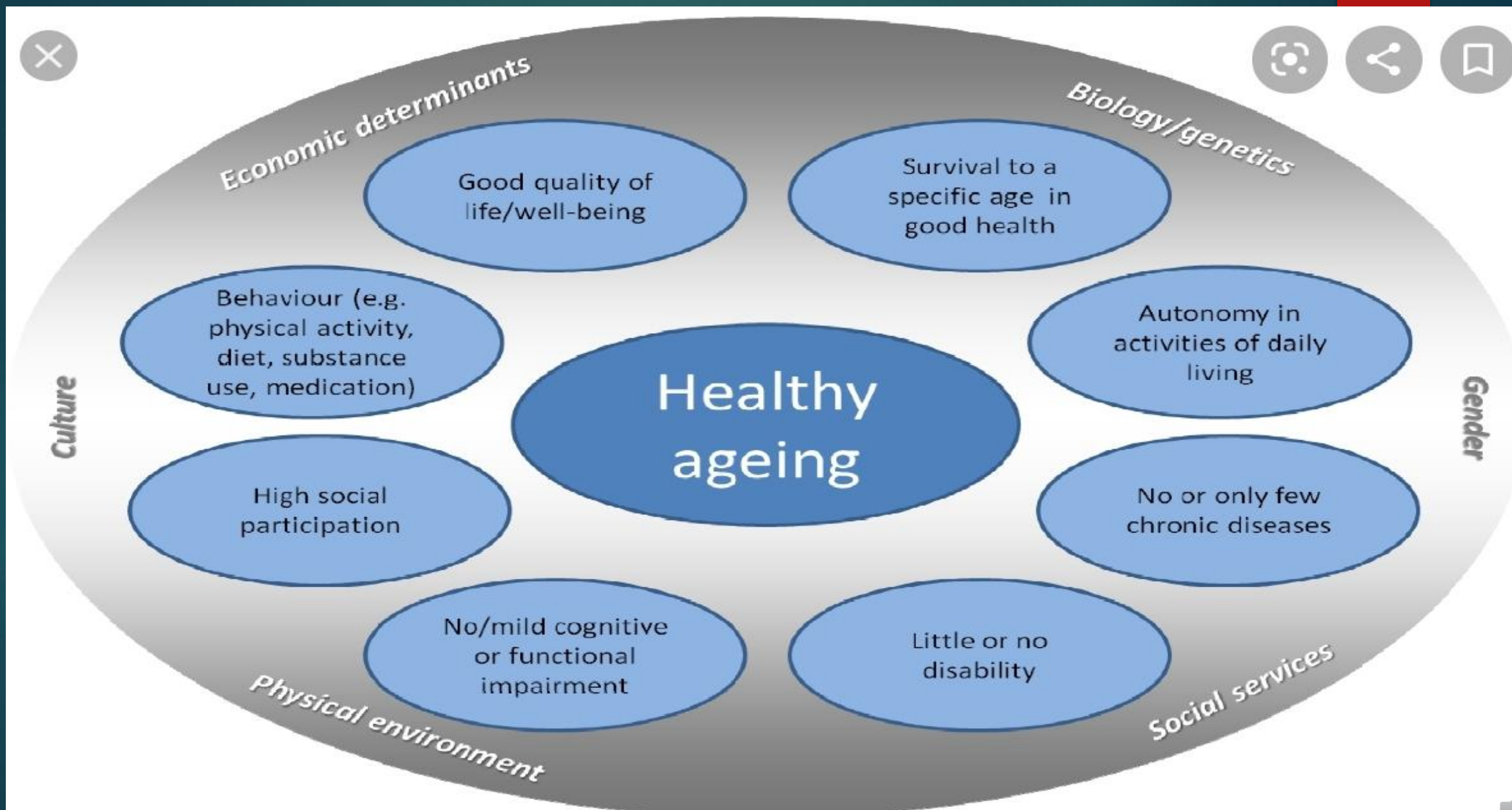


Demographics of aging

► According to the WHO, the global population is increasing at the annual rate of 1.7%, while the population of those over **65 years is increasing at a rate of 2.5%**. Both the developed, as well as the lesser-developed countries, are expected to experience significant shifts in the age distribution of the population by 2050. The fastest growing population segment in most countries is the adults older than 80 years, which according to the United Nations estimates will make up nearly 20% of the world's population.^{1,2}

► The proportion of people in **Saudi Arabia aged 60 or more is predicted to be 25 percent of the total population of 40 million by the end of 2050**. Moreover, the number of people aged 80 or more is expected to reach 1.6 million, or 4 percent of the total population in the same period.





Trends in oral health in older adults

- ▶ Oral health in population is improving
- ▶ Studies have indicated that rates of periodontal disease and caries have decreased for most age groups
- ▶ Contributing Factors:
- ▶ Introduction of **fluoridation** through community water treatment, fluoridated tooth paste, mouth rinse, professionally applied fluoride gels, advances in dental treatment modalities, **improved oral hygiene** and **access to dental services**
- ▶ **Edentulism** (complete tooth loss) is one of the most important indicator of oral health. It affects quality of life, self esteem and nutritional status.
- ▶ In economically developed countries, the trend of edentulism has **declined**



Common Behavioral Oral-Systemic Disease Risk Factors in geriatric patients

- ▶ **Diet:** Consumption of high saturated fats and carbohydrates, and low intake of fiber, fruits, and vegetables, as well as sedentary lifestyle practices
- ▶ **Smoking:** related to periodontal disease and poor oral health/hygiene, and contributes to the development of 30% of all cancer
- ▶ **Stress:** Elevated levels of salivary cortisol have been linked to extent and severity of periodontal disease and temporomandibular joint dysfunction that may exacerbate chronic systemic conditions, frailty, and demise.
- ▶ **Alcohol :** Consumption has been associated with traumatic injuries; chronic use will lead to neurological, cognitive impairment and increased risk of suicide.
- ▶ **Hygiene:** Effective removal of the oral microbiome has been associated with good oral health and reducing risk of chronic inflammatory systemic conditions
- ▶ **Physical inactivity:** increase the risk of multiple chronic systemic conditions, such as falls, cardiovascular disease, diabetes, obesity, frailty, and osteoporosis, and Alzheimer's



Effect of aging and oral health

- ▶ **Functional status and oral health**
- ▶ increased risk of physical disability
- ▶ ability to maintain good oral hygiene and restrict their access to necessary dental care
- ▶ financial resources: dental insurance ceases upon retirement

- ▶ **Systemic Diseases**
- ▶ Diabetes, Hypertension
- ▶ Cardiovascular disease
- ▶ **Medications and oral health**
- ▶ Saliva provides a crucial role in oral health.
- ▶ Xerostomia (dry mouth)
- ▶ Medications - Antihypertensives, Anticoagulants, Antidepressants

- ▶ **Cognitive function and oral health**
- ▶ Alzheimer's disease (AD) or other dementias
- ▶ not view dental care as a high priority and may have limited self-awareness of dental care needs, decline in oral hygiene practices,



Effect of aging and oral health

Physiological Changes

- ▶ ↓ Healing
- ▶ ↓ Immunity
- ▶ ↓ Saliva
- ▶ Polypharmacy {multiple drugs}

These changes affect treatment outcomes.

Common Oral Conditions seen in geriatric patients



Root caries

Gum disease

Xerostomia

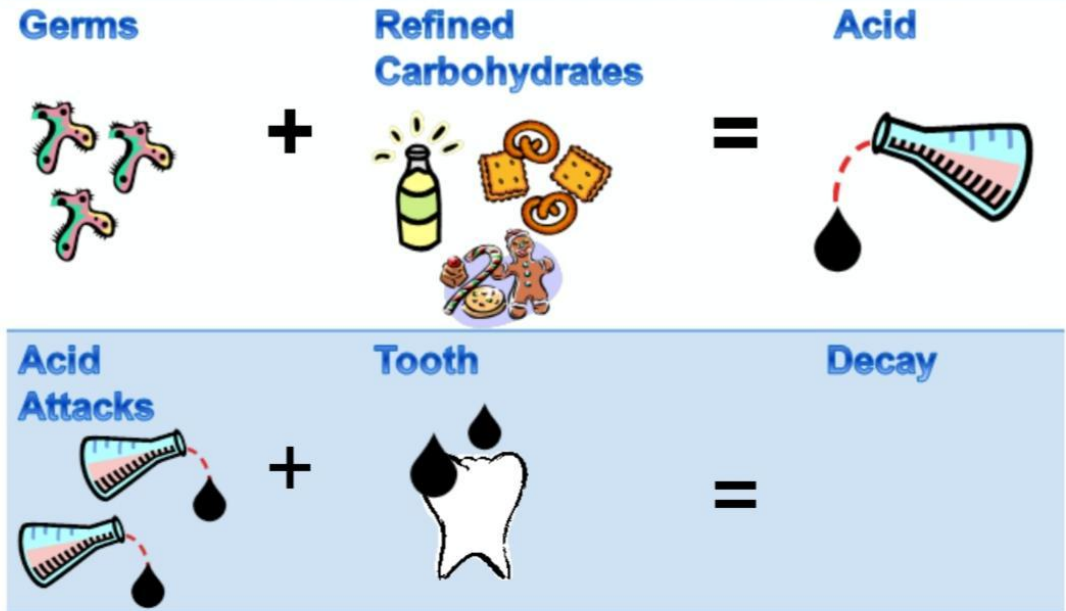
Oral cancer

Edentulism

Root caries

- ▶ major cause of tooth loss in older adults
- ▶ Dental caries (tooth decay) is caused by specific bacteria (*Streptococcus mutans*, *Streptococcus sobrinus*, lactobacilli, and others) that colonize tooth surfaces, feed on carbohydrates, and produce acids as waste products. These acids dissolve the mineral content of the tooth, and if not halted or reversed, a carious lesion (cavity) is formed.
- ▶ The root comprises the biologic structures cementum and dentin. Root surface cementum and dentin are more susceptible to cavitation because they are less mineralized than enamel

Process of Tooth Decay



Acid Attacks Last for 20 Minutes



Root caries

- ▶ Treating root caries can be technically challenging.
- ▶ The location of the root caries may be difficult to access; it often may extend below the gingival margin, making it necessary to retract the gingiva with a clamp, pack retraction cord to expose the cervical margin of the lesion, or utilize laser or electrosurgery to recontour the gingiva and obtain access to the lesion
- ▶ Restorative materials: amalgam, composite, glass ionomer
- ▶ Amalgam: greater longevity, lower costs
- ▶ Resin bonding to dentin or enamel requires adequate isolation and saliva contamination control. This is time consuming and often difficult to achieve in restoring root caries lesions at or near the gingival margin
- ▶ Glass ionomer, resin-modified glass ionomer, contain fluoride and release it into the saliva and adjacent tooth structure. (Atraumatic restorative treatment)



Oral Disease



- **Gum Disease**

- **Gingivitis** – inflammation of the gum tissue
- **Periodontitis** – destruction of bone and the structures supporting the teeth
 - Common oral problems that may result in tooth loss
 - Respond well to proper oral hygiene routines and treatment by your dentist

- Periodontal disease is an inflammatory disease caused by gram-negative anaerobic bacteria from dental plaque
- Periodontal disease progresses to periodontitis when the inflammation extends to the periodontal ligament and alveolar bone which lead to loss or recession of gingival tissue, decrease in alveolar bone mass, tooth mobility/tooth loss, and potentially edentulism



Periodontal Disease and Diabetes

People with diabetes and uncontrolled blood sugar...

- Are more likely to have periodontal disease
- May have more difficulty controlling blood sugar
- Are more susceptible to contracting infections
- Should be treated to eliminate perio infection



Xerostomia (dry mouth)

- ▶ is a subjective sensation of oral dryness and it may be associated with salivary gland hypofunction (SGH), and changes in salivary composition.
- ▶ It is a very common complaint among older adults but it is not a normal part of the aging process
- ▶ **Xero= Dry, Stomia= mouth**



Dry mouth

Unstimulated Flow (resting salivary flow—no external stimulus)

- Typically 0.2 mL – 0.3 mL per minute
- Less than 0.1 mL per minute means the person has hyposalivation
 - Hyposalivation – not producing enough saliva



NORMAL FUNCTION OF SALIVA³

- Hydrating–moisturizing
- Cleansing
- Lubrication
- Digestion
- Remineralization of dentition
(pH maintenance, buffering)
- Maintenance of mucosal integrity
- Immunity mediator
- Antimicrobial (antifungal, antibacterial)
- Stimulation of minor salivary glands
- Cellular maintenance
- Enables swallowing
- Enables tasting
- Enables speech articulation



XEROSTOMIA: Epidemiology

Factors that Affect Salivary Flow

- Medication
- Autoimmune disease (Sjogren's syndrome, lupus)³
- Systemic diseases (diabetes, asthma, kidney, sarcoidosis, HIV)³
- Stress/anxiety/depression⁵
- Radiation therapy to the head and neck³
 - 30 Gy = glandular fibrosis (gland can still produce some saliva)
 - 60-70 Gy = glandular destruction (gland can no longer produce saliva)
- Gender (70 % female, usually postmenopausal³)
- Sympathomimetic medications (stimulate the sympathetic nervous system)⁶
- Parasympatholytic medications (inhibit the parasympathetic nervous system)⁶
- Circadian rhythms (decreases in the fall and increases in the spring)



XEROSTOMIA: Diagnosis

Symptoms^{3,5}

- Viscous saliva
- Sticky saliva
- Difficulty speaking
- Difficulty swallowing
- Halitosis
- Altered taste
- Complaint of dryness
- Complaint of burning mouth, lips, or tongue
- Altered sense of smell



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Signs

- Increased caries
- Food sticking to the oral structures
- Frothy saliva
- Gingivitis
- Absence of saliva
- Cracking and fissuring of the tongue
- Ulceration of oral mucosa

XEROSTOMIA: Management

Treatment of Xerostomia-Associated Problems¹⁴

Dental Caries

- Daily use of fluorinated dentifrice (0.05-percent sodium fluoride)
- Daily use of prescription fluoride gel (1.0-percent sodium fluoride, 0.4-percent stannous fluoride)
- Application of 0.5-percent sodium fluoride varnish to teeth
- Dental examinations at least every 6 months and bitewing radiographs every 12 months for early diagnosis

Dry Mouth

- Oral moisturizers/lubricants, mouthwashes, and sprays
- Sugar-free gums, mints, lozenges
- Artificial salivary replacements
- Prescription sialogogues: pilocarpine (5 mg 3 times per day and at bedtime); cevimeline (30 mg 3 times per day)
- Lubricants on lips every 2 hours
- Use of bedside humidifier during sleeping hours

Dysgeusia

- Drinking of fluids while eating

Dysphagia

- Careful eating, with fluids
- Copious use of fluids during meals
- Avoidance of dry, hard, sticky, and difficult-to-masticate foods



Xerostomia Management

Identify Cause



Drug-induced? → Modify medication (consult physician)



Stimulate saliva → chewing gum



Saliva substitutes



Fluoride therapy



Regular follow-up

Oral cancer

Cancer

- Can develop in the mouth
- More likely to develop in older adults who smoke or chew tobacco and/or abuse alcohol
- Early diagnosis is crucial
- All non-healing sores in the mouth or changes in voice deserve close attention and monitoring

Painless in the early stages

Report symptoms

- Sores that do not heal within 2 weeks
- Lumps, bumps, or swelling; red or white patches
- Difficulty chewing or swallowing

Edentulism- Denture Patient



- ▶ Process of accepting and using CD is complex
- ▶ Requires adaptation of learning, muscular skill and motivation
- ▶ Pt's ability and willingness to accept and learn to use dentures ultimately determine the success of clinical treatment
- ▶ Pt interview and clinical examination , identification of anticipated difficulties keys to successful treatment
- ▶ Effective communication – develops trusting



Mental Health / Attitude

- House's Classification of Patients

Type of patient	Attitude	Principal Characteristics	Prognosis
Philosophical	trusting	Accepts advise	good
Exacting / critical	doubting	Gives advise to surgeon	Fair/poor
Hysterical / Skeptical	demanding	Unpleasant past experience	poor
Indifferent	unconcerned	Sent by relatives	fair

Oral Lesion	Clinical Presentation	Treatment
Burning Mouth Syndrome (BMS)	Persistent burning sensation in the mouth, often without visible lesions. May be associated with dry mouth or taste changes.	No definitive cure; Clonazepam, capsaicin, lidocaine, and supportive therapy.
Oral Candidiasis (Thrush)	White, curd-like plaques on the tongue, palate, and buccal mucosa that can be wiped off, leaving a red, raw surface.	Antifungal medications (nystatin, fluconazole, clotrimazole).
Lichen Planus	White, lacy patches (Wickham's striae) or painful erosions on the mucosa.	Corticosteroids (topical or systemic), immunosuppressants.
Xerostomia (Dry Mouth)	Dryness of the oral mucosa, difficulty swallowing, burning sensation, increased risk of cavities and infections. 	Saliva substitutes, sialogogues (pilocarpine), hydration, and good oral hygiene.

Leukoplakia	White patches that cannot be scraped off, often on the tongue or buccal mucosa; may be pre-malignant.	Regular monitoring, biopsy if suspicious, tobacco cessation.
Erythroplakia	Red, velvety patches in the oral mucosa, high risk of malignancy.	Biopsy, close follow-up, surgical excision if needed.
Oral Squamous Cell Carcinoma (OSCC)	Non-healing ulcers, red/white patches, pain, bleeding, or difficulty swallowing.	Surgery, radiation, chemotherapy, depending on stage.
Denture Stomatitis	Red, inflamed areas under dentures, often due to Candida infection.	Antifungals, proper denture hygiene, removal of dentures at night.
Angular Cheilitis	Cracks, redness, and soreness at the corners of the mouth, often due to fungal or bacterial infection.	Antifungal or antibacterial creams, vitamin supplementation (if deficient).
Traumatic Ulcers	Painful ulcers due to mechanical irritation from dentures, sharp teeth, or accidental biting.	Removal of cause, topical analgesics, healing within 1-2 weeks.

Hairy Tongue	Elongated filiform papillae on the dorsal tongue, often black or brown in color due to bacterial overgrowth and staining.	Improved oral hygiene, tongue scraping, discontinuation of contributing factors (e.g., smoking, certain medications).
Geographic Tongue	Irregular, smooth, red patches on the tongue with white borders that change location over time. May cause mild discomfort.	Usually self-limiting; symptomatic treatment with topical steroids if needed. Avoid spicy/irritating foods.
Herpes Zoster (Shingles) in the Oral Cavity	Painful, unilateral vesicular eruptions along the trigeminal nerve distribution, may crust over and cause postherpetic neuralgia.	Antiviral therapy (acyclovir, valacyclovir) within 72 hours, pain management, supportive care.
Epulis Fissuratum	Overgrowth of fibrous tissue along the edges of an ill-fitting denture, appearing as folds of tissue.	Adjustment or replacement of denture, surgical excision if severe.

Planning for dental treatment in the older adult

Patient Evaluation



Medical History + Drug History



Functional Status Assessment



Risk Assessment (Systemic + Oral)



Define Treatment Priority



Modify Treatment Plan



Preventive + Maintenance Phase

Importance of Communication

Geriatric pt may have Hearing impairment
,Cognitive decline

- ▶ Speak slowly
- ▶ Use visual aids
- ▶ Involve caregiver /Family member

- ▶ **Ethical Considerations**
 - Informed consent
 - Elder abuse awareness

Treatment Planning Principles

- ▶ Modify, don't deny
- ▶ Short appointments
- ▶ Morning visits
- ▶ Stress reduction
- ▶ Patient comfort is key.
- ▶ **Focus on Preventive Strategies**
- ▶ Fluoride therapy
- ▶ Oral hygiene
- ▶ Diet counseling
- ▶ Recall visits

**Maintaining healthy teeth and gums
at any age is an important part
of preserving your overall good health.**



Conclusion



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